INTRO TO R

Tengku Muhammad Hanis Bin Tengku Mokhtar, PhD

March 10, 2026

About me

Senior Lecturer
at **UniSZA**

Background

- PhD (Public Health Epidemiology) from USM, 2024
- MSc (Medical Statistics) from USM, 2019
- MBBCh from Al-Azhar University, 2015

Interest:

- Medical statistics, meta-analysis, bibliometrics
- Machine learning and deep learning application in medical sciences
- Application of R on health/medical data

Contact me:

- Email: tengkuhanismokhtar@gmail.com
  tengkuhanismokhtar@unisza.edu.my
- Website: https://tengkuhanis.netlify.app/
  https://jomresearch.netlify.app/

Material

https://github.com/tengku-hanis/ml_rg_unisza.git

Learning outcome

- Not to know everything about machine learning and R
- To know enough so that you can further explore and learn on your own

Content

- Why use R
- R, RStudio, and Posit
- Installation
- Data types in R
- Data structures in R
- Indexing and data wrangling
- Core packages in tidyverse
- Missing values and duplicates in R
- Basic plots
- Tips

Why use R

- Flexibility to use R everywhere - R is free
- Wide functionalities:
  - Statistical analysis
  - Machine learning, deep learning
  - Image processing (to certain extent)
  - Website development
  - Etc
- Wide range of packages (for example, in CRAN - 23,351 packages):
  - Wide range of statistical analysis: basic to advanced
  - Early access to new statistical method
  - All packages are free

One letter that can change your life, forever.

R, RStudio, and Posit

- R:
  - Programming and statistical software
  - Run in the back of IDE
- RStudio:
  - Integrated development environment (IDE)
  - Run on front, which the user interact with
- Posit:
  - Previously, also known as RStudio
  - The company that build RStudio

- Other IDEs for R:
  - Google Collab
  - PyCharm
  - VS code - Visual Studio Code
  - Positron
  - And many more
- Why use RStudio:
  - Supported by Posit
  - Probably one of the most (if not the most) popular IDE for R
  - Large support community

Installation

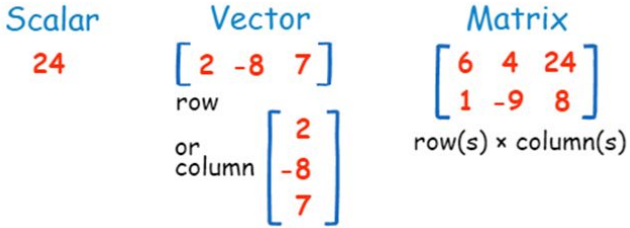
- Posit Cloud (we will be using this through out this workshop):
  - Cloud version of RStudio
  - No need installation
  - Free up to an extent
  - Need to make an account
- RStudio:
  - Install base R first
  - Install RStudio

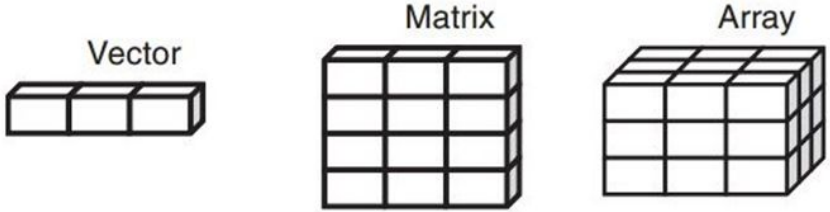
Data types in R

- Most common data types in R:
  - Character: “hanis”, “school”
  - Numeric: 1.0, 2.1, 3.0, 3.2
  - Integer: 1, 2, 3, 4L
  - Logical: TRUE, FALSE
  - Complex:
    - Example: 1+4i
    - Not commonly used
    - Used to store number and an imaginary components

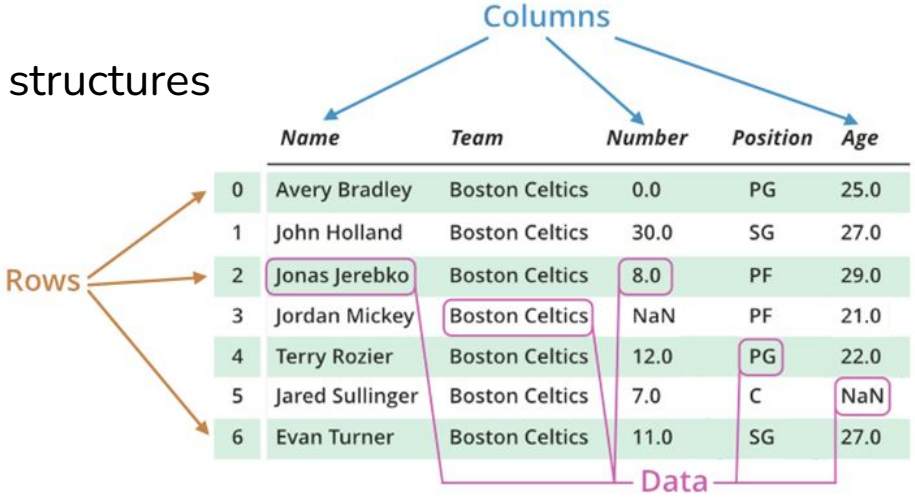
Data structures in R

- Most common data structure in R:
  - Atomic vector:
    - Basic data structure in R
    - Can contain one data type at a time
  - Matrix
  - Array:
    - Matrix but with more than one layer
    - More common in image analysis





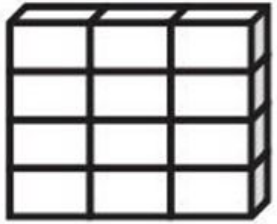
- Data frame
  - Contains the names of variable in the column names and each observation in row
  - Each column may contain different data type
- List
  - May contains different data structures



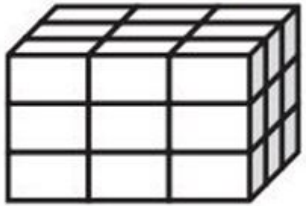
	Name	Team	Number	Position	Age
0	Avery Bradley	Boston Celtics	0.0	PG	25.0
1	John Holland	Boston Celtics	30.0	SG	27.0
2	Jonas Jerebko	Boston Celtics	8.0	PF	29.0
3	Jordan Mickey	Boston Celtics	NaN	PF	21.0
4	Terry Rozier	Boston Celtics	12.0	PG	22.0
5	Jared Sullinger	Boston Celtics	7.0	C	NaN
6	Evan Turner	Boston Celtics	11.0	SG	27.0

Vector

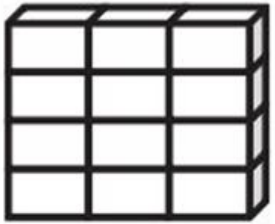
Matrix



Array



Data frame



Columns can be different modes

List

Vectors

Arrays

Data frames

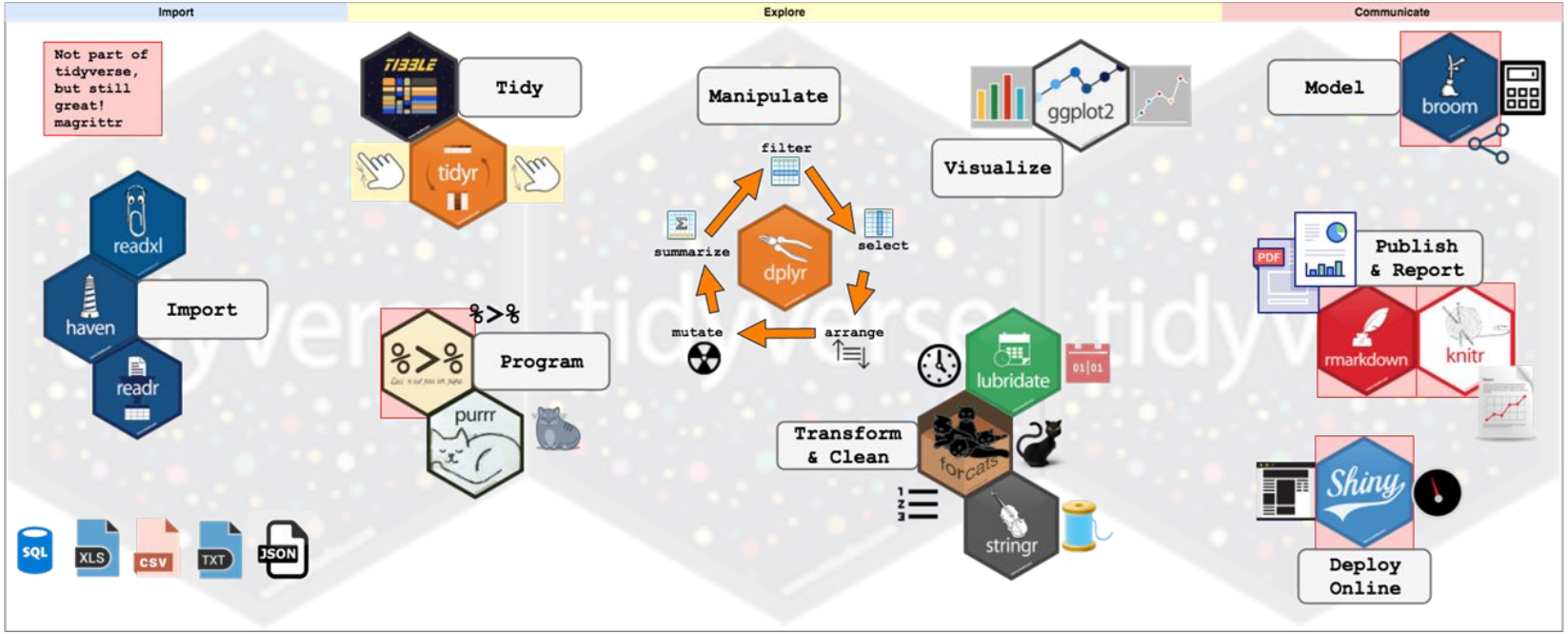
Lists

Indexing and data wrangling

- Indexing:
  - Selecting certain elements using their indices
  - This is most commonly done using base R
- Data wrangling:
  - Getting the data into a useful form for visualization, summary, and modeling
  - Can be done using base R and tidyverse approach
  - Tidyverse:
    - A collection of R packages design for data science - developed by Posit team
    - Total of 30 packages with 8 core packages
    - Quite popular among R user especially beginners

Core packages in tidyverse

Some common packages in tidyverse



Missing values and duplicates in R

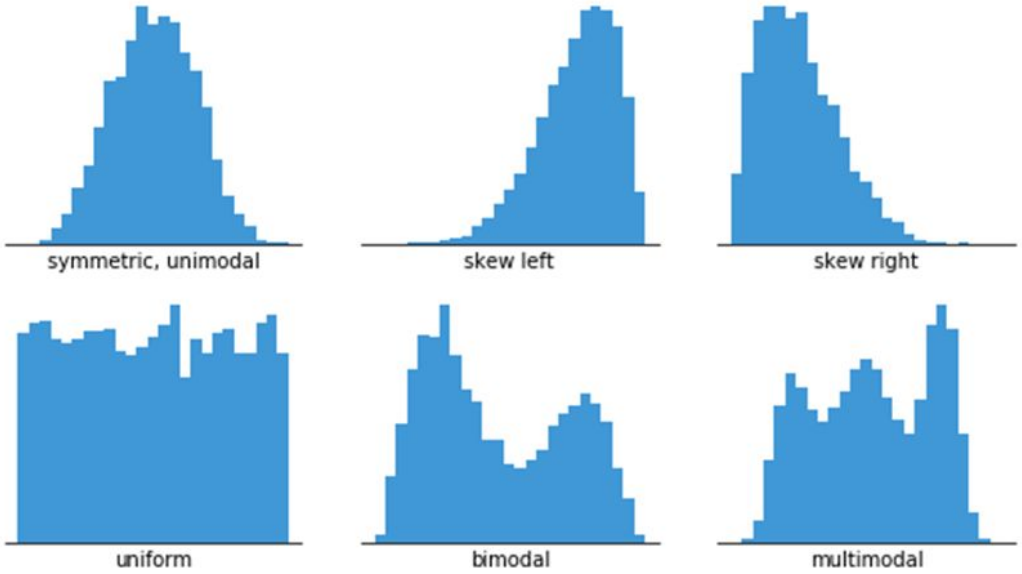
- Missing values:
  - Missing value represented by NA (not available)
  - Impossible value represented by NaN (not a number)
  - NA and NaN is not the same as “NA” and “NaN”
- Duplicates:
  - To identify duplicates in R, a unique value is needed such as ID number, IC number, hospital ID, etc

Basic plots

- Histogram
- Bar plot
- Box plot
- Scatter plot

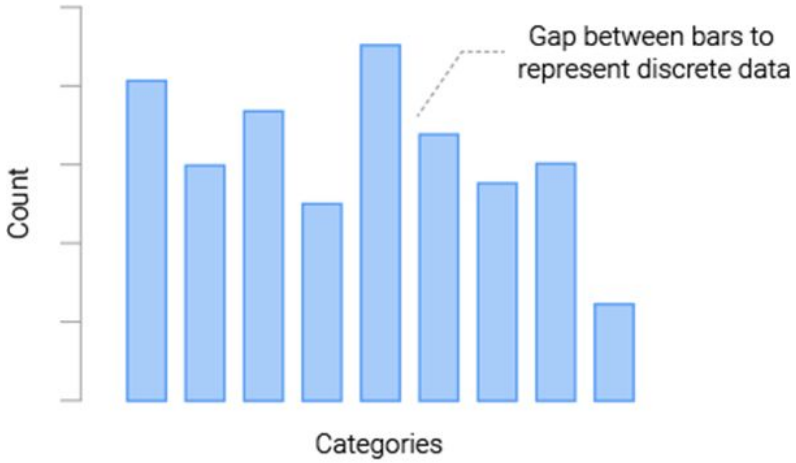
Histogram

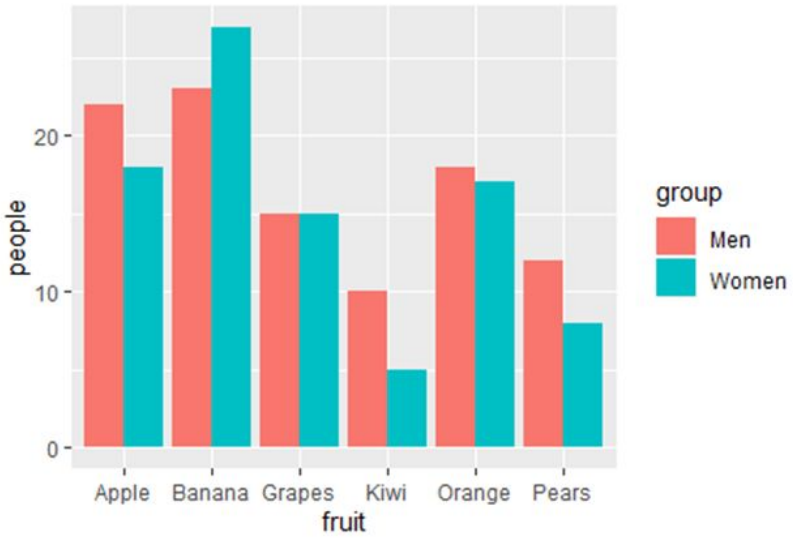
- Describe the distribution of the numeric values



Bar plot

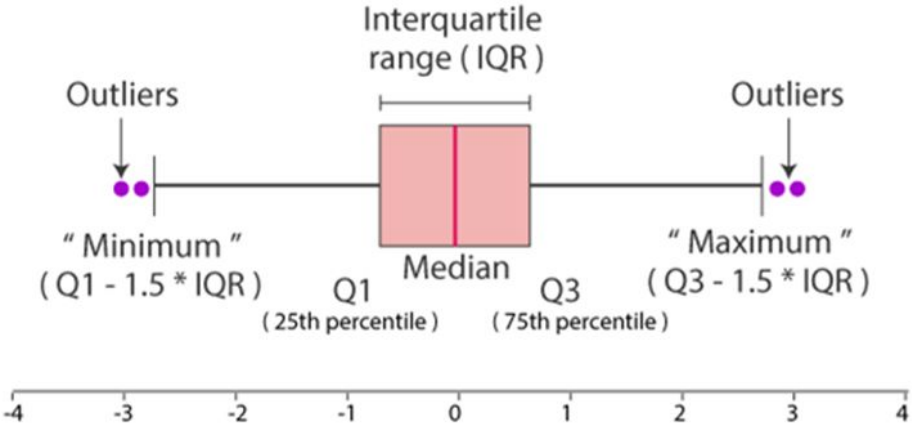
- Describe the distribution of the categorical variable





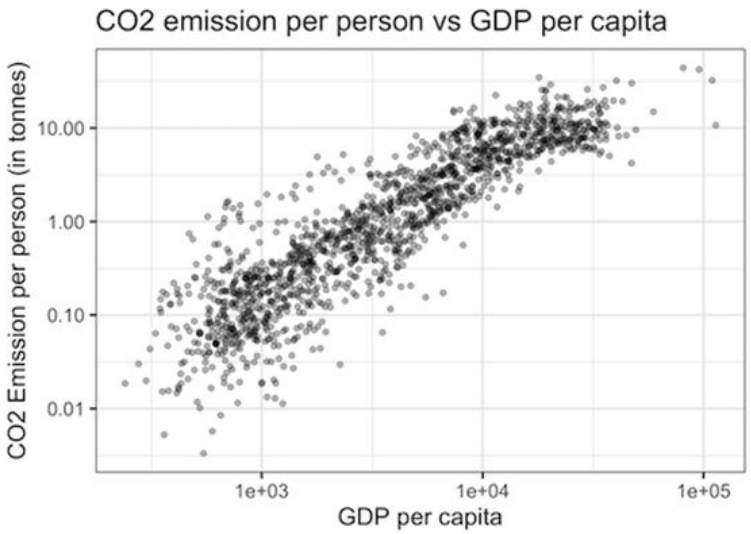
Box plot

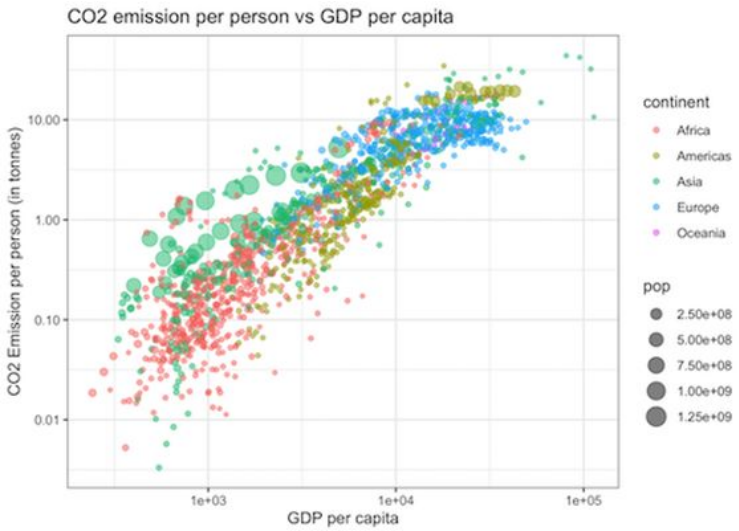
- Also known as box and whisker plot
- Can be used to detect outliers - values that deviate too much from other values in the datasets



Scatter plot

- Represent values for two numeric variables





Tips

- It takes some time before you going to be comfortable and familiar with R
- The more you use R, the faster you become familiar with R

Suggested readings/references

- Hanis, T. M. (2026, January 15). R essentials: A beginner's Guide to Data Analysis. https://tengku-hanis.github.io/r_essential/

Links

- Exercise for R basics - https://tengku-hanis.github.io/intro2r_answers/
- Install R and RStudio on Mac OS - https://www.youtube.com/watch?v=1PsPfMaLWSk&t=46s
- Install R and RStudio on Windows - https://www.youtube.com/watch?v=h2IPWVXaUuU

Any question?

https://tengkuhanis.netlify.app/
https://jomresearch.netlify.app/